1.125 Agentic Engineering

When Should Forecasts Trust Context?

TS-Cov: A research proposal for agent-guided context selection in time-series forecasting

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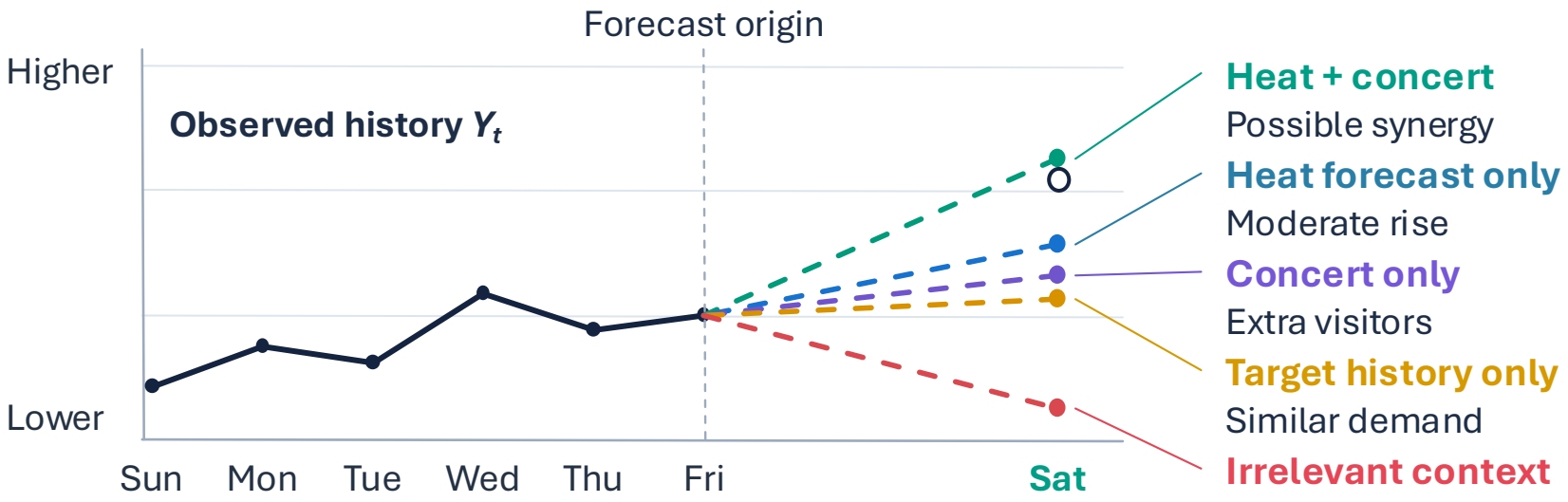
October 29, 2026

Which information should a forecaster use?

City electricity load at 6 PM

Plan power generation for Saturday at 6 PM.

Illustrative 6 PM load



Actual (later)

Heat + concert
Possible synergy

Heat forecast only
Moderate rise

Concert only
Extra visitors

Target history only
Similar demand

Irrelevant context
False outage rumor

Heat + crowds may amplify cooling demand.

Available context

Known before forecast

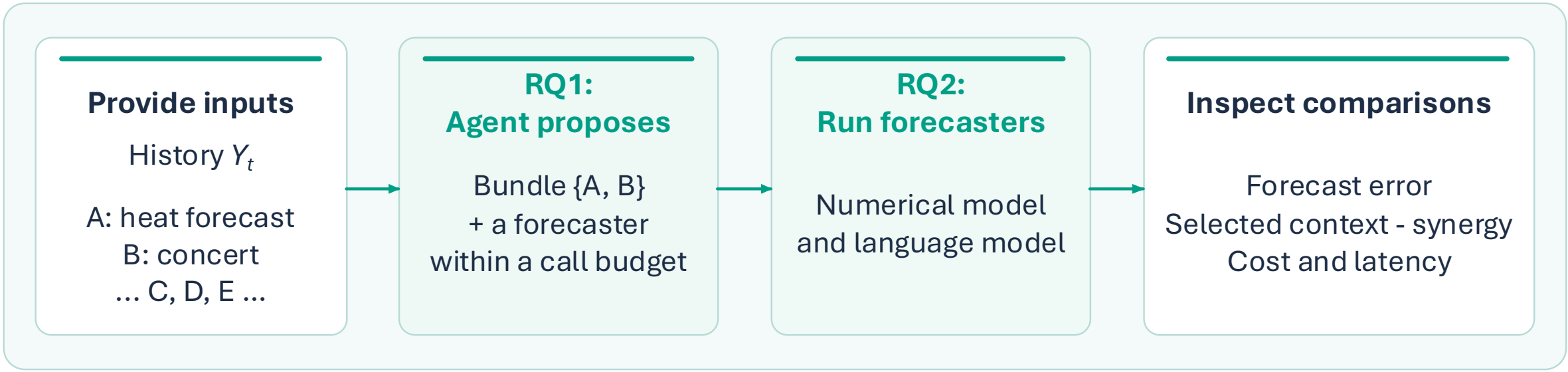
- $X_{1,t}$ Weather forecast: 36°C
- $X_{2,t}$ Calendar: Saturday
- $X_{3,t}$ News: tomorrow's tariff is set
- $X_{4,t}$ Event: major concert scheduled

Can an agent choose useful context bundles and suitable forecasters better than fixed rules?

Users: forecasting researchers and operations analysts

Proposed solution

A reproducible forecasting experiment



Approach

Candidate data, models and evaluation

Candidate data

A small **TimesX** subset of daily commodity and currency series with contextual information.

Models and resources

TSFM: numerical inputs and covariates
LLM: history with selected text

Common comparison protocol

Baselines	No context, all context, greedy selection
Evaluation	Develop on earlier periods. Test on untouched later periods. Compare forecast error (MASE)

TimesX: Liu et al. (2026), arXiv:2607.06973

Chronos-2: Ansari et al. (2025), arXiv:2510.15821

A focused first study with manageable risks

Core scope

- One dataset, a small task panel,
- Two forecasters (LLM and TSFM)
- One budgeted agent strategy.

Why this project

This is a research direction
I want to continue pursuing.

The course can provide a
focused starting point for
reproducible follow-up work.

Risks and planned responses

Cost and latency

Cap search rounds and API calls.
Cache repeated evaluations.

Future information leakage

Audit context timestamps and model vintage.
Disclose remaining leakage risks.

Don’t have a desired performance and no product

Use strong baselines and multiple test windows.
Report negative findings and failure cases.

A completed, reproducible investigation

Milestones

After feedback
Confirm data and protocol.
Implement baseline runs.

Midpoint
Demonstrate a working pipeline
and one budgeted agent search.

Final
Complete held-out comparisons.
Release code, README and findings.

Success criteria

- Reproduce a run from the repository.
- Use the same candidate inputs and test periods.
- Report error, cost, latency and failure cases.
- Reach a clear conclusion, even if agents do not help.

Final demonstration Replay one case, inspect the context and forecast, then show the benchmark comparison.